

SQL MASTER TREE

1. Database Fundamentals

- What is DB / DBMS / RDBMS
- Tables, Rows, Columns
- Primary Key
- Foreign Key
- Candidate Key
- Composite Key
- **Constraints**
 - NOT NULL
 - UNIQUE
 - PRIMARY KEY
 - FOREIGN KEY
 - CHECK
 - DEFAULT
- Data Integrity

2. SQL Data Types

- **Numeric**
 - INT
 - BIGINT
 - DECIMAL
 - FLOAT
- **String**
 - CHAR
 - VARCHAR
 - TEXT
- **Date & Time**
 - DATE
 - TIME
 - DATETIME
 - TIMESTAMP
- Boolean / Binary

3. DDL

(Data Definition Language)

- **CREATE**
 - DATABASE
 - TABLE
 - INDEX
- **ALTER**
 - ADD COLUMN
 - MODIFY COLUMN
 - DROP COLUMN
- **DROP**
 - DATABASE
 - TABLE
- TRUNCATE

4. DML

(Data Manipulation Language)

- INSERT
- UPDATE
- DELETE
- MERGE / UPSERT

5. DQL

(Data Query Language)

- SELECT
- DISTINCT
- **WHERE**
 - AND
 - OR
 - NOT
- ORDER BY
- GROUP BY
- HAVING
- LIMIT / OFFSET

6. SQL Operators

- Arithmetic (+ – * /)
- Comparison
- = != < > <= >=
- Logical (AND OR NOT)
- BETWEEN
- IN
- LIKE
- IS NULL

7. SQL Functions

- **Aggregate**
 - COUNT
 - SUM
 - AVG
 - MIN
 - MAX
- **String**
 - CONCAT
 - SUBSTRING
 - LENGTH
 - TRIM
- **Numeric**
 - ROUND
 - ABS
- **Date**
 - NOW
 - DATEADD
 - DATEDIFF

8. Joins

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- CROSS JOIN
- SELF JOIN

9. Subqueries

- Scalar Subquery
- Correlated Subquery
- Nested Subquery

10. Views

- CREATE VIEW
- UPDATE VIEW
- MATERIALIZED VIEW

11. Indexing

- Clustered Index
- Non-Clustered Index
- Composite Index
- Index Optimization

12. Transactions

- BEGIN
- COMMIT
- ROLLBACK
- SAVEPOINT

13. ACID Properties

- Atomicity
- Consistency
- Isolation
- Durability

14. Normalization

- 1NF
- 2NF
- 3NF
- BCNF
- Denormalization

15. Advanced SQL

- Stored Procedures
- Triggers
- CTE (WITH)
- **Window Functions**
 - ROW_NUMBER
 - RANK
 - DENSE_RANK
 - PARTITION BY
- Recursive Queries